Case 09: Mixed-Mode Ignition-Delay Effects

1 Verification purpose

This case verifies how firebrand ignition delay alters spread when surface propagation and spotting operate together. The surface-fire-dominated transport simulation condition delays firebrand development until the surface front controls advance; combined surface-fire and firebrand transport permits sufficiently distant deposits to develop ahead of that front.

The paired design exposes incorrect delay application, a missing transition between propagation modes, or an inconsistent surface-rate reference. Success requires surface-controlled behavior in the first simulation condition and quantified earlier arrival, spot ignition, and leading-edge gain in the second.

2 Mathematical formulation and numerical implementation

2.1 Surface-fire and firebrand time scales

The verification configuration is a flat, uniform fire behavior fuel model (FBFM) 102 grassland with wind speed $U_w = 6.71\,\mathrm{m\,s^{-1}}$. The surface head-fire spread rate U_s is read from the surface-dominated simulation condition spread rate raster produced by the Rothermel model in the tested ELMFIRE executable. For the reported run, $U_s = 0.577\,\mathrm{m\,s^{-1}}$. In the absence of a successful firebrand-induced ignition, the exact one-dimensional solution is

$$t_s(x) = \frac{x}{U_s}, \qquad x_{\mathrm{LE},s}(t) = U_s t, \qquad \overline{\mathrm{ROS}}_{\mathrm{LE},s} = U_s. \tag{1}$$

For a firebrand emitted at x_s and transported a downwind distance X, the earliest time at which it can produce a fully developed fire is

$$t_{\mathrm{spot}}(x_s, X) = \frac{x_s}{U_s} + \frac{X}{U_w} + t_{\mathrm{ign,small}} + t_{\mathrm{ign,large}}. \tag{2}$$

The surface front reaches the same target location at $t_s(x_s + X) = (x_s + X)/U_s$. With immediate smoldering-to-flaming transition, a spot fire can lead the surface front only when

$$X\left(\frac{1}{U_s} - \frac{1}{U_w}\right) > t_{\mathrm{ign,large}}. \tag{3}$$

2.2 ELMFIRE implementation exercised

The current implementation uses the non-superseded spotting path with these explicit selectors:

- generation proportional to fire power;
- an empirical transport-distance distribution; and
- Eulerian accumulation.

For each burning cell, the firebrand-generation calculation evaluates

$$\dot{N}_{\mathrm{cell}} = \frac{I_f\,\Delta y}{1000}\,GR^*, \tag{4}$$

where I_f is fireline intensity in kW m^{-1}, Δy is the cell width, and $GR^* = 33.3\,\mathrm{s}^{-1}\,\mathrm{MW}^{-1}$. The factor $I_f \Delta y/1000$ is the heat-release rate of the cell in MW. The Eulerian transport calculation deposits these particles according to the configured Sardoy-type distribution ($\mu = 2.18$, $\sigma = 1.23$), truncated at an effective maximum distance of approximately 150 m.

The ignition calculation first converts the deposited number to a number per unit area. In the surface-dominated simulation condition, the physical ignition model applies the wildland mass-load criterion before sampling SFT, with $\vartheta_{\text{ign,small}} = 42\,\mathrm{s}$ and $t_{\text{ign,large}} = 300\,\mathrm{s}$. For this parameter set, the documented limiting behavior predicts that deposited mass remains below the physical threshold, so the SFT probability is zero [1]. In the mixed-mode simulation condition, the simple ignition model prescribes immediate SFT and a 150 s development delay. When the development delay expires, the level-set state, arrival-time field, and ember-ignition diagnostic are updated by the same firebrand ignition event.

3 Derivation of expected behavior

At the maximum transport distance $X_{\text{max}} = 150\,\mathrm{m}$, flight takes

$$t_f = \frac{150}{6.71} = 22.4\,\mathrm{s}, \qquad t_s(150) = \frac{150}{0.577} = 260.0\,\mathrm{s}.$$

For surface-dominated simulation condition, even the deliberately optimistic assumption of immediate SFT would give $t_f + t_{\text{ign,large}} = 322.4\,\mathrm{s}$, later than the surface arrival. The expected result is therefore Equation 1, with no cells attributed to firebrand ignition.

For mixed-mode simulation condition, the earliest full spot fire at X_{max} occurs at $22.4 + 150 = 172.4\,\mathrm{s}$. Moreover, the left side of Equation 3 is 237.6 s, which exceeds the 150 s development delay. Long-distance deposits can therefore mature ahead of the primary front. The expected signature is an acceleration near 170 s, at least one firebrand-ignited cell, a leading edge ahead of $U_s t$, and a mean spread rate greater than U_s over the common analysis interval. These are the qualitative features expected from mixed-mode spread.

The lead condition can also be solved for the minimum useful spotting distance:

$$X_{\text{crit}} = \frac{t_{\text{ign,large}}}{1/U_s - 1/U_w}, \qquad \frac{1}{0.577} - \frac{1}{6.71} = 1.584071\ \mathrm{s\,m}^{-1}.$$

For the surface-dominated 300-s delay, $X_{\text{crit}} = 300/1.584071 = 189.39$ m, which exceeds the configured $X_{\text{max}} = 150$ m; even the longest deposit cannot lead. For the mixed 150-s delay, $X_{\text{crit}} = 94.69\,\mathrm{m} < 150$ m, so the tail $94.69 < X \le 150$ m can mature ahead of the surface front. At $X = 150$ m,

$$t_f = 150/6.71 = 22.3547\ \mathrm{s}, \quad t_s = 150/0.577 = 259.9653\ \mathrm{s}, \quad t_s - t_f = 237.6106\ \mathrm{s}.$$

Thus the mixed earliest full-spot scale is $22.3547 + 150 = 172.3547$ s, whereas the surface-dominated optimistic scale is $22.3547 + 300 = 322.3547\ \mathrm{s} > t_s$. These substitutions yield the explicit reference signatures used by the metrics: zero surface-dominated ember ignitions, positive mixed-mode ember ignitions, mixed $ROS/U_s \ge 1.10$, at least 20 m edge and paired-time of arrival (TOA) advantages, and onset in the 120–220 s interval around 172.35 s.

4 Verification metrics and acceptance criteria

The postprocessor excludes the two-cell halo and operates on the centre-row TOA and spread-rate profiles. It converts the positive median Rothermel spread rate value from ft/min to m/s and uses it as U_s throughout the reference calculations. It fits $t = a + x/U$ over $50\,\mathrm{m} \le x \le 280\,\mathrm{m}$ using at least 12 cells. The following independently informative conditions must all pass:

1. surface-dominated simulation condition mean TOA relative error against x/U_s is at most 0.10.
2. surface-dominated simulation condition fitted rate of spread (ROS) relative error against U_s is at most 0.10.
3. surface-dominated simulation condition contains no firebrand-ignited physical cells.
4. mixed-mode simulation condition contains at least one firebrand-ignited physical cell.
5. mixed-mode simulation condition fitted ROS is at least $1.10U_s$.
6. At 250 s, its leading edge is at least 20 m ahead of $U_s t$.
7. Its first one-cell leading-edge gain occurs from 120 to 220 s, bracketing the derived 172.4 s scale.
8. Its median TOA is at least 20 s earlier than surface-dominated simulation condition over the shared fitting interval.

Missing outputs do not pass by default. Both simulation conditions must provide final TOA, accumulated-firebrand, and ember-ignition GeoTIFFs before any case-level pass decision is possible.

4.1 Justification of metric quantification

The two 10% surface-baseline limits allow cell-centred TOA and regression quantization while requiring agreement with the run-derived Rothermel speed over a 230 m fitting interval, more than twenty grid cells. The zero spot count in the surface-dominated simulation condition and the positive count in the mixed simulation condition are exact mechanism checks and therefore have no numerical tolerance. For the mixed mode, $ROS/U_s \geq 1.10$ requires a 10% effect beyond the baseline fitting band, while the 20 m edge and paired-TOA leads equal two spatial cells and are too large to arise from one-cell front classification alone.

The 120–220 s onset window brackets the independently derived 172.4 s earliest full-spot scale by approximately ± 50 s. That width accommodates stochastic deposition, cell crossing, and the fact that the metric detects the first full-cell gain rather than a continuous event, but still rejects acceleration unrelated to the predicted delay regime. The eight behavior checks are conjunctive because a faster fitted front without ember attribution, timing consistency, and a paired TOA lead would not verify firebrand-delay coupling.

4.2 Predeclared acceptance criteria

Table 1 summarizes the metrics fixed before inspection of the calculated results. Numerical values and component statuses are reported separately in Section 7.

Table 1: Predeclared verification metrics and acceptance criteria.

Metric	Output, region, and aggregation	Acceptance criterion
Output completeness	Final TOA, accumulated-firebrand, and ember-ignition rasters for both current simulation conditions; halo and no-data excluded.	2 of 2 simulation conditions
Surface-dominated TOA error	Mean relative error against x/U_s on 50–280 m; at least 12 cells.	≤ 0.10
Surface-dominated ROS error	Relative error of fitted ROS against run-derived Rothermel U_s.	≤ 0.10
Surface-dominated spot count	Number of ember-ignited physical cells.	0
Mixed-mode spot count	Number of ember-ignited physical cells.	≥ 1
Mixed-mode ROS ratio	Fitted mixed-mode ROS divided by U_s.	≥ 1.10
Mixed-mode edge gain	Leading-edge distance ahead of $U_s t$ at 250 s.	≥ 20 m
Acceleration onset	First time the leading edge exceeds the surface reference by one cell.	120–220 s
Paired TOA lead	Median surface-dominated minus mixed-mode TOA over common fitting cells.	≥ 20 s
Overall decision	Conjunction of completeness and all eight behavior checks.	PASS only if all pass

5 Simulation configuration and predicted outcome

Figure 1 records the prepared spatial inputs for combined surface-fire and firebrand transport. These panels show the prepared spatial fields, remove the two-cell numerical buffer, and retain map coordinates in metres. The left panel shows the most informative fuel or structure layout available for the case; the right panel shows the cells selected by the non-positive initial level-set field, making a localized ignition or firebrand-emission source explicit.

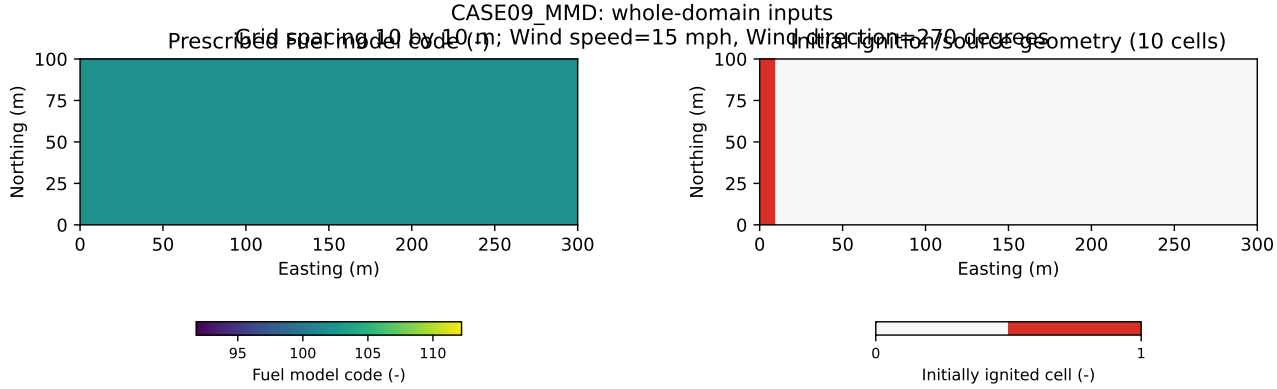


Figure 1: Whole-domain prepared input configuration for the representative simulation condition. The physical domain is shown without the numerical halo; coordinates, configured wind values, fuel or structure layout, and initial ignition/source geometry are identified in the panels.

The Courant–Friedrichs–Lewy (CFL) number measures the distance advected during one timestep relative to the grid spacing and constrains the time discretization.

Table 2: Parameters common to both verification simulation conditions.

Parameter	Value	Verification role
Physical domain	300×100 m	one-dimensional centre transect
Raster domain	34×14 cells	includes two-cell halo
Cell size	10 m	controlled baseline resolution
Time step / wind CFL	0.742 574 s / 0.4979	common exact-duration divisor
Fuel and terrain	FBFM 102, flat	homogeneous surface spread
Wind	6.71 m s^{-1} toward $+x$	firebrand transport
Ignition geometry	full crosswind line at $x = 5$ m	1-D behavior
Generation	PER-MW, $GR^* = 33.3$	heat-release-scaled source
Transport / accumulation	empirical / Eulerian	reference transport path
Surface fire	enabled	general coupled fire spread

Table 3: Ignition parameters changed by the paired experiment.

Simulation condition	Ignition model	$t_{\text{ign,small}}$	$t_{\text{ign,large}}$	Prediction
surface-dominated simulation condition	PHYSICAL	42 s scale	300 s	surface dominated
mixed-mode simulation condition	SIMPLE	0 s	150 s	mixed-mode acceleration

5.1 Preprocessing, execution, and postprocessing

The two simulation conditions use independently prepared, co-registered spatial fields. Both represent flat terrain, fuel model 102, and constant wind; places a line ignition across every physical crosswind row; and writes explicit model selectors and ignition parameters into each local ELMFIRE simulation configuration. The halo is present numerically but is excluded from all reported quantities. The nominal 0.5-wind-CFL step is 0.745645 s. Because the controlled stops are 300 and 450 s, their greatest common duration is 150 s; preprocessing selects $N = \lceil 150/0.745645 \rceil = 202$ and the common $\Delta t = 150/202 = 0.742574$ s. Both stops are therefore exact timestep multiples and the actual wind CFL remains below 0.5.

Both prescribed conditions are simulated independently with ELMFIRE. Their final physical fields are compared using the surface-rate reference and the arrival-time, ignition, and leading-edge measures defined above.

6 Actual simulation results

Figure 2 shows the representative time-of-arrival field from combined surface-fire and firebrand transport. The displayed field is taken from the simulated results, masks missing values, and excludes the two-cell numerical buffer. The domain map complements the spread-rate summaries by showing the combined surface-fire and delayed-firebrand arrival pattern.

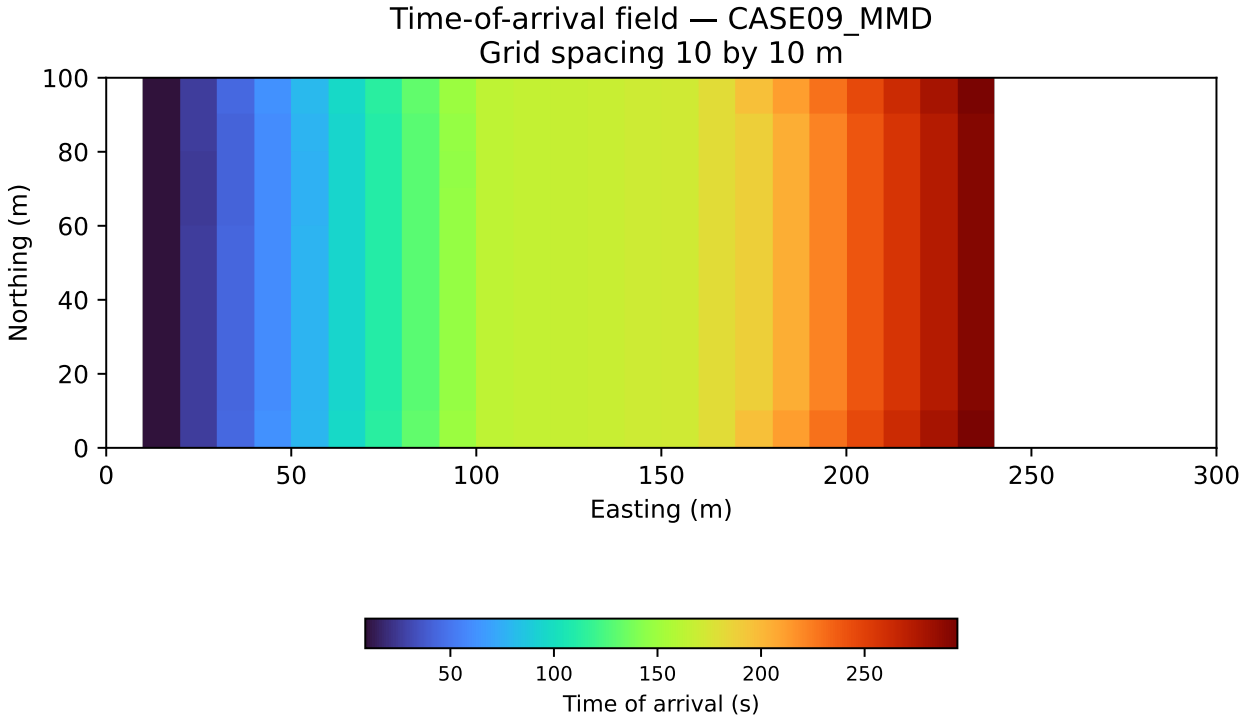


Figure 2: Whole-domain time-of-arrival field from the representative completed ELMFIRE run. This domain-scale result supplements, rather than replaces, the higher-level verification profiles, convergence plots, ensemble statistics, and calculated acceptance metrics below.

Figures 3 and 4 first present each ignition regime independently using the same four observables: accumulated firebrands, arrival time, leading-edge position, and leading-edge ROS. Colored curves are actual ELMFIRE outputs; black curves are the Rothermel surface-spread references.

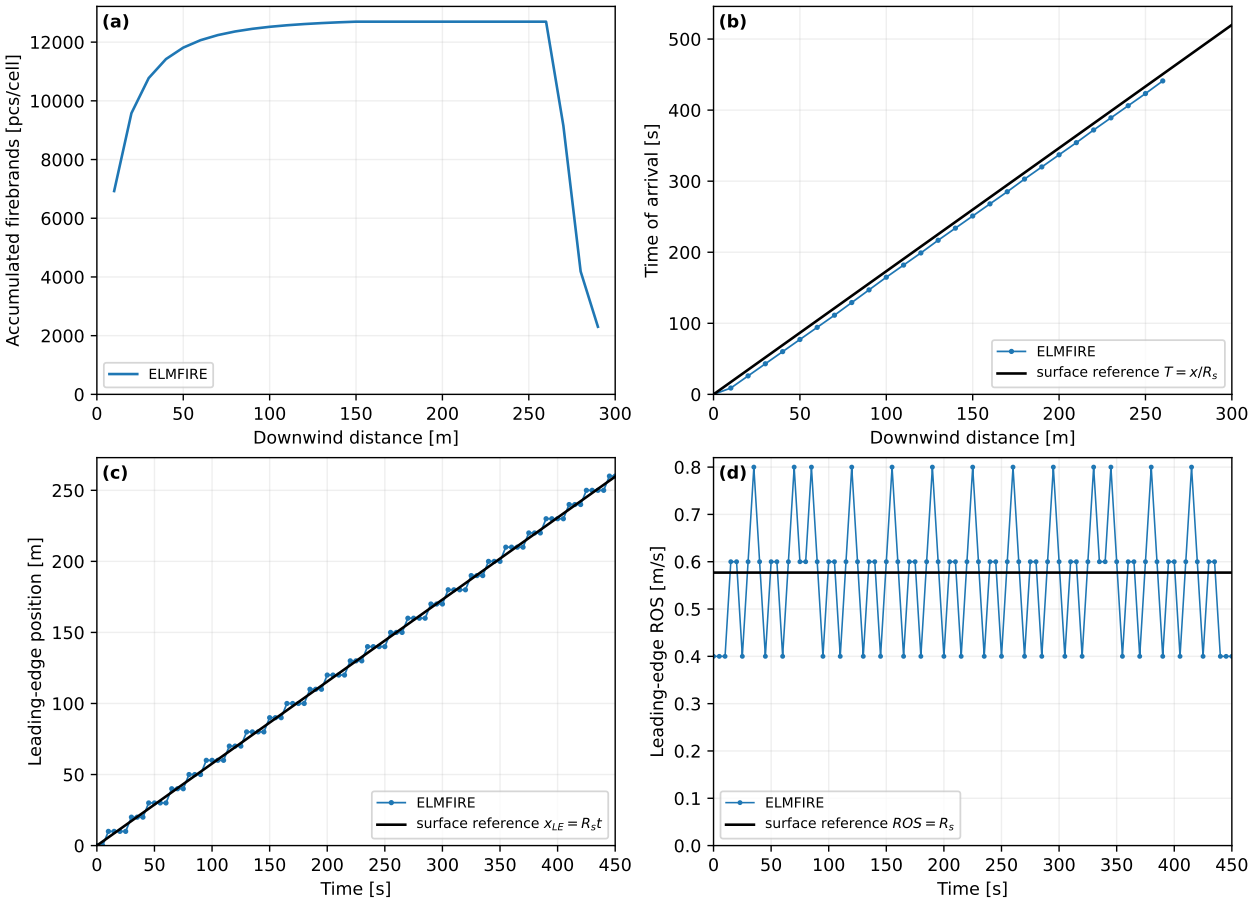


Figure 3: Long-delay surface-dominated simulation condition. The leading edge remains close to the surface-spread characteristic despite continued firebrand deposition.

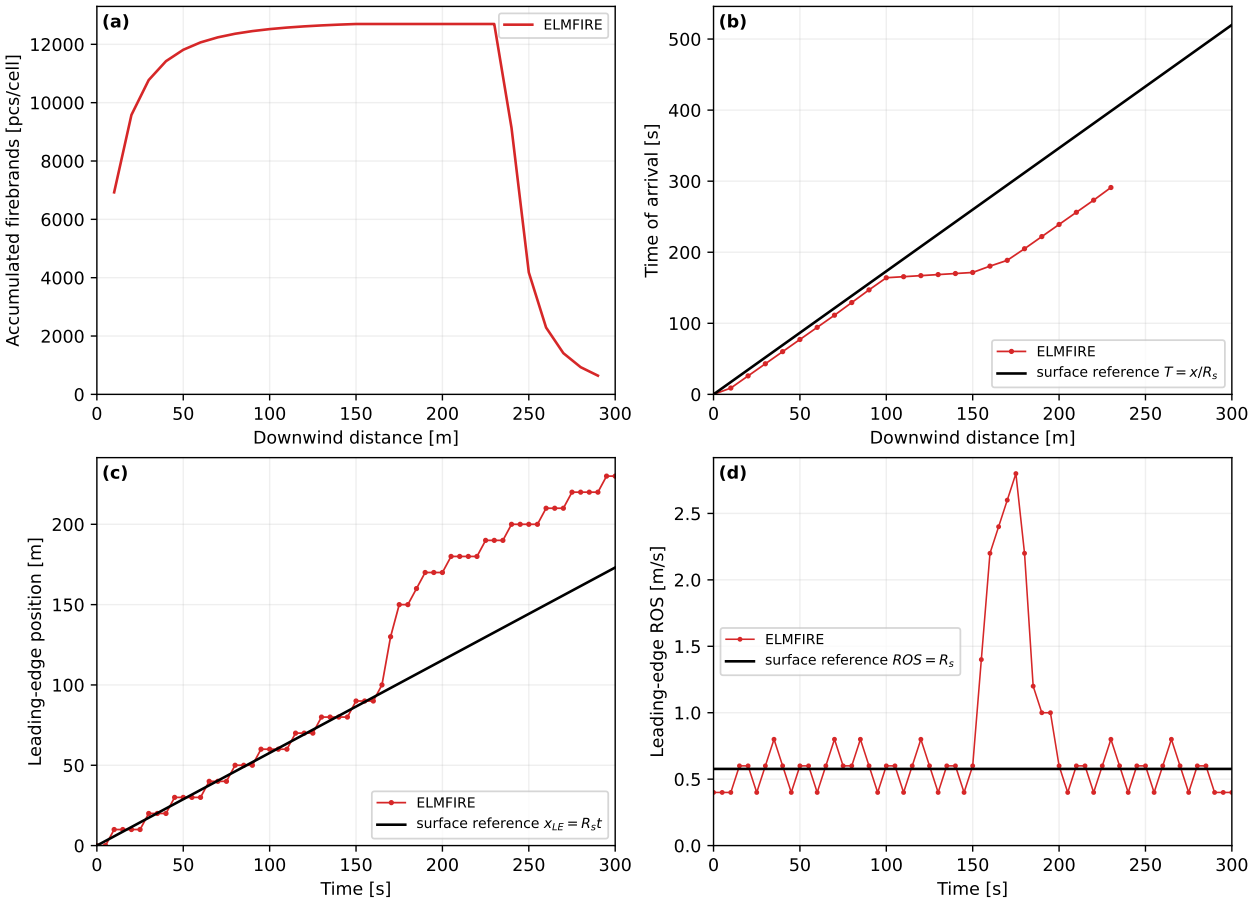


Figure 4: Finite-delay mixed-mode simulation condition. Departure from the surface-only arrival, trajectory, and ROS references identifies firebrand-induced advance.

Figure 5 overlays the two simulation conditions as a supplemental controlled comparison, making the timing and magnitude of their divergence easier to identify.

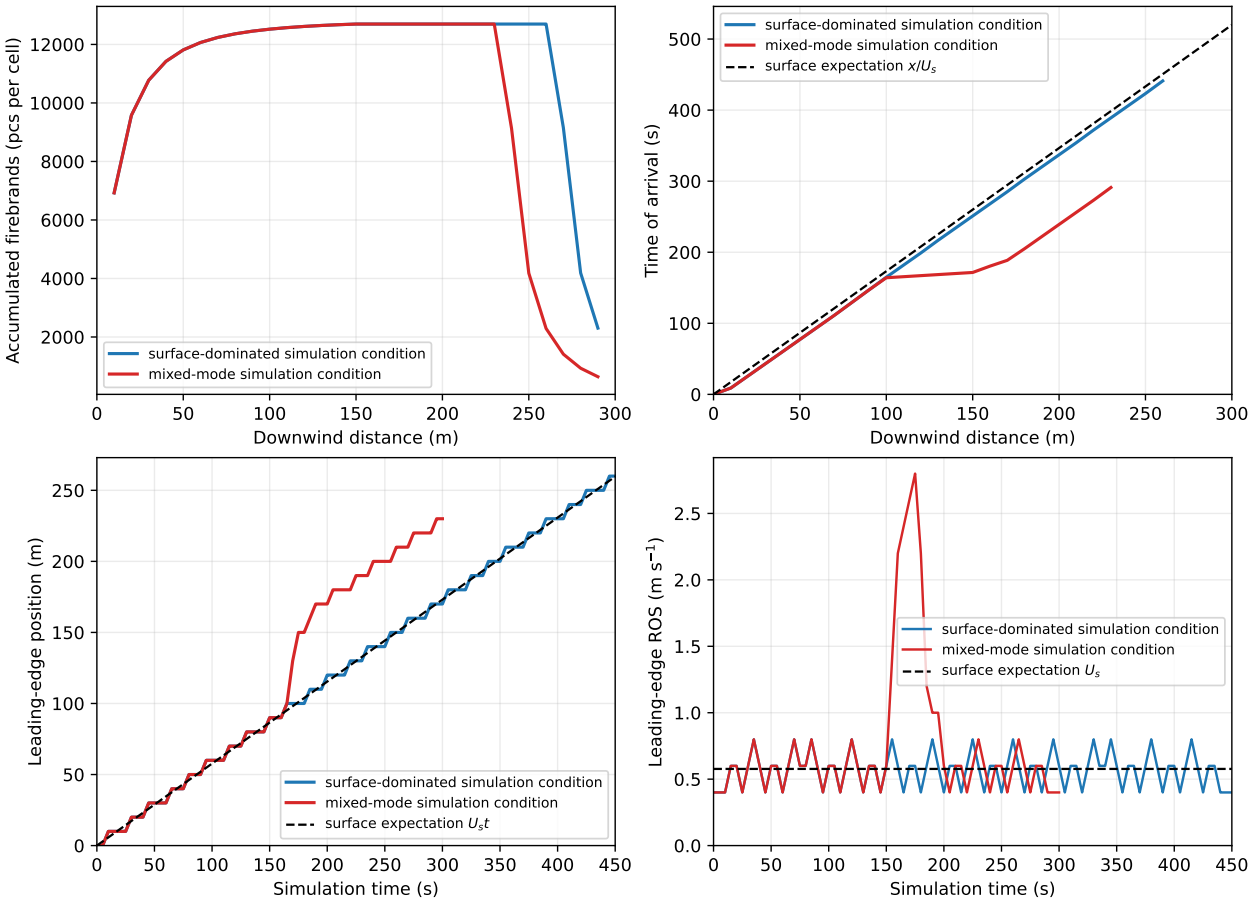


Figure 5: Paired ELMFIRE comparison of the long-delay surface-dominated and finite-delay mixed-mode regimes. Accumulation is crosswind averaged; the other panels use the centre-row arrival-time field.

7 Calculated metrics and verification decision

Table 4 reads the calculated entries directly from calculated verification and validation metrics. An empty simulation result is represented as N/A with NOT EVALUABLE, never as an empty table cell or an implicit pass.

Table 4: Acceptance metrics calculated from the paired ELMFIRE outputs.

Metric	Acceptance limit	Calculated	Status
Output completeness	2 simulation conditions	2/2	PASS
surface-dominated simulation condition mean TOA error	≤ 0.10	0.043	PASS
surface-dominated simulation condition mean ROS (m s^{-1})	$0.577 \pm 10\%$	0.578	PASS
surface-dominated simulation condition ember-ignited cells	≤ 0	0	PASS
mixed-mode simulation condition ROS / U_s	≥ 1.10	1.743	PASS
mixed-mode simulation condition ember-ignited cells	≥ 1	64	PASS
mixed-mode simulation condition edge gain at 250 s (m)	≥ 20	55.746	PASS
mixed-mode simulation condition acceleration onset (s)	120–220	170.0	PASS
Median mixed-mode simulation condition TOA lead (s)	≥ 20	63.861	PASS
Overall decision	all components pass	PASS	PASS

8 Technical interpretation

The current paired execution passes every component. The surface-dominated mean ROS is 0.578 m/s with no ember-ignited cells. The mixed-mode mean ROS is 1.006 m/s (1.743 times the surface reference), with 64 ember-ignited cells, a 55.746-m leading-edge gain, acceleration onset at 170.0 s, and a median TOA lead of 63.861 s.

A pass demonstrates two complementary properties: delayed firebrand states do not spuriously outrun the surface solver in the surface-dominated limiting case, and reducing the delay activates a resolvable acceleration in the mixed-mode simulation condition. If surface-dominated simulation condition fails alone, inspect the physical mass threshold, development-state timing, and arrival-time update. If mixed-mode simulation condition fails alone, inspect the simple ignition probability, persistence of deposited ember states across level-set steps, delayed transition to a full ignition, and the coupling that writes that ignition to the level set.

This is numerical verification of the controlled formulation, not field validation. The ELMFIRE experiment is two-dimensional but deliberately uses a uniform crosswind line so the centre transect approximates the corresponding one-dimensional calculation. The mixed-mode thresholds diagnose the expected mixed regime rather than an unavailable closed-form mixed-mode solution. Stochastic ignition means the exact onset may vary, which is why its acceptance window brackets rather than equals 172.4 s.

The current analysis note is: All metrics were calculated from the two actual ELMFIRE simulation condition outputs; the surface reference is the Rothermel ROS read from the surface-dominated simulation condition VS raster.

9 Scope and limitations

This paired test isolates the interaction between surface propagation and firebrand ignition delay in a uniform domain. It verifies implementation behavior and timing attribution; it does not validate either spread mechanism against field observations.

References

[1] Y. Qin, *A Physics-Based Eulerian Framework for Modeling Firebrand Showering in Regional-Scale Wildland and Wildland–Urban Interface Fire Simulations*, Ph.D. dissertation, University of Maryland, College Park, 2025.